Name: ..... Due: **Next Class / Friday**

C3 Early Multitrack Recordings HW 3 – Jimi Hendrix 'Purple Haze' & Studio Innovation

Artist: The Jimi Hendrix Experience (Producer: Chas Chandler, Engineer: Eddie Kramer)
https://www.youtube.com/watch?v=WGoDaYjdfSg

1. In 1967, Eddie Kramer mixed "Purple Haze" at Olympic Studios. Identify the hard-panning characteristics of this classic 1960s multitrack stereo mix. (3)

Track Element	Stereo Pan Position
Bass Guitar (Noel Redding)	Centre / Right
Rhythm Guitar & Fuzz Overdubs	
Drums (Mitch Mitchell)	
Lead Vocal (Jimi Hendrix)	Centre

2. What distinctive non-standard guitar harmonic interval (dissonance) is played in the opening riff from 0'00" to 0'08"? (1)

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3. Why did early stereo multitrack mixes between 1964 and 1968 feature extreme "hard-panning" (e.g. instruments hard-left, vocals hard-right) rather than smoothly variable panning? (2)

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4. Ken Townsend invented Automatic Double Tracking (ADT) in 1966. Explain how ADT works using tape delay and a Variable Speed Oscillator (VSO) to create a natural doubled effect without re-recording. (4)

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5. Hendrix popularized extreme analog distortion and filtering effects. Explain how the Dallas-Arbiter Fuzz Face (germanium transistors) and the Vox Wah-Wah pedal modify audio waveforms. (4)

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6. Describe how you would close-mic a loud 4x12 Marshall valve guitar cabinet in a modern studio to capture high-gain rock tones without overloading the preamp: (4)

Choice of Microphone & Polar Pattern (2)
Microphone (Dynamic / Ribbon): Polar Pattern:
Placement, Angle & Distance relative to Speaker Cone / Cap Edge (2)

Total:	**/18**